Analytical Laboratory Report

Report ID: S60442.01(01)
Generated on 04/23/2024

Report to

Attention: Cindy Tomaszewski
City of Cadillac - WWTP
1121 Plett Road
Cadillac, MI 49601

Phone: 231-775-2368 FAX:
Email: lab@cadillac-mi.net

Report produced by

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Report Summary

Lab Sample ID(s): S60442.01
Project: Qtrly Biosolids
Collected Date(s): 04/01/2024
Submitted Date/Time: 04/02/2024 10:00
Sampled by: David Skiff
P.O. #:

Table of Contents

Maya Murshak
Technical Director

Analytical Laboratory Report

General Report Notes

Analytical results relate only to the samples tested, in the condition received by the laboratory.
Methods may be modified for improved performance.
Results reported on a dry weight basis where applicable.
'Not detected' indicates that parameter was not found at a level equal to or greater than the reporting limit (RL).
When MDL results are provided, then 'Not detected' indicates that parameter was not found at a level equal to or greater than the MDL.
40 CFR Part 136 Table II Required Containers, Preservation Techniques and Holding Times for the Clean Water Act specify that samples for acrolein and acrylonitrile, and 2-chloroethylvinyl ether need to be preserved at a pH in the range of 4 to 5 or if not preserved, analyzed within 3 days of sampling.
QA/QC corresponding to this analytical report is a separate document with the same Merit ID reference and is available upon request.
Starred (*) analytes are not NY NELAP accredited.
Samples are held by the lab for 30 days from the final report date unless a written request to hold longer is provided by the client.
Report shall not be reproduced except in full, without the written approval of Merit Laboratories, Inc.
Limits for drinking water samples, are listed as the MCL Limits (Maximum Contaminant Level Concentrations)
PFAS requirement: Section 9.3.8 of U.S. EPA Method 537.1 states "If the method analyte(s) found in the Field Sample is present in the FRB at a concentration greater than 1/3 the MRL, then all samples collected with that FRB are invalid and must be recollected and reanalyzed."
Samples submitted without an accompanying FRB may not be acceptable for compliance purposes.
Wisconsin PFAs analysis: MDL = LOD; RL = LOQ. LOD and LOQ are adjusted for dilution.
All accreditations/certifications held by this laboratory are listed on page 3. Not all accreditations/certifications are applicable to this report.
For a specific list of accredited analytes, please feel free to contact the laboratory or visit https://www.meritlabs.com/certifications.

Report Narrative

There is no additional narrative for this analytical report

Analytical Laboratory Report

Laboratory Accreditations (For Reference Only)

Authority	Accreditation ID
Michigan DEQ	#9956
DOD ELAP & ISO/IEC 17025:2017	#69699 PJLA Testing
WBENC	#2005110032
Ohio VAP	#CL0002
Indiana DOH	#C-MI-07
New York NELAC	#11814
North Carolina DENR	#680
North Carolina DOH	#26702
Pennsylvania DEP	#68-05884
Wisconsin DNR	FID# 399147320

Qualifier Descriptions

Qualifier	Description
!	Result is outside of stated limit criteria
B	Compound also found in associated method blank
E	Concentration exceeds calibration range
F	Analysis run outside of holding time
G	Estimated result due to extraction run outside of holding time
H	Sample submitted and run outside of holding time
I	Matrix interference with internal standard
J	Estimated value less than reporting limit, but greater than MDL
L	Elevated reporting limit due to low sample amount
M	Result reported to MDL not RDL
O	Analysis performed by outside laboratory. See attached report.
R	Preliminary result
S	Surrogate recovery outside of control limits
T	No correction for total solids
X	Elevated reporting limit due to matrix interference
Y	Elevated reporting limit due to high target concentration
b	Value detected less than reporting limit, but greater than MDL
e	Reported value estimated due to interference
j	Analyte also found in associated method blank
o	Associated EIS outside of control limits
p	Benzo(b)Fluoranthene and Benzo(k)Fluoranthene integrated as one peak.
q	Qualifier ion ratio outside of control limits
x	Preserved from bulk sample

Glossary of Abbreviations

Abbreviation	Description
RL/RDL	Reporting Limit
MDL	Method Detection Limit
MS	Matrix Spike
MSD	Matrix Spike Duplicate
SW	EPA SW 846 (Soil and Wastewater) Methods
E	EPA Methods
SM	Standard Methods
LN	Linear
BR	Branched

Method Summary

Method	Version
ASTM D7968-17M	ASTM Method D7968 - 17 Modified (Isotopic Dilution)
SM2540B	Standard Method 2540 B 2015

Parameter Summary

Parameter	Synonym	Cas #
PFBA	Perfluorobutanoic Acid	375-22-4
PFPeA	Perfluoropentanoic Acid	2706-90-3
4:2 FTSA	4:2 Fluorotelomer Sulfonic Acid	757124-72-4
PFHxA	Perfluorohexanoic Acid	307-24-4
PFBS	Perfluorobutane sulfonic Acid	375-73-5
PFHpA	Perfluoroheptanoic Acid	375-85-9
PFPeS	Perfluoropentane Sulfonic Acid	2706-91-4
6:2 FTSA	6:2 Fluorotelomer Sulfonic Acid	27619-97-2
PFOA	Perfluorooctanoic Acid	335-67-1
PFHxS	Perfluorohexane Sulfonic Acid	355-46-4
PFHxS-LN	Perfluorohexane Sulfonic Acid - LN	355-46-4-LN
PFHxS-BR	Perfluorohexane Sulfonic Acid - BR	355-46-4-BR
PFNA	Perfluorononanoic Acid	375-95-1
8:2 FTSA	8:2 Fluorotelomer Sulfonic Acid	39108-34-4
PFHpS	Perfluoroheptane Sulfonic Acid	375-92-8
PFDA	Perfluorodecanoic Acid	335-76-2
N-MeFOSAA	N-methyl perfluorooctanesulfonamidoacetic acid	2355-31-9
EtFOSAA	N-Ethyl Perfluorooctane Sulfonamidoacetic Acid	2991-50-6
PFOS	Perfluorooctane Sulfonic Acid	1763-23-1
PFOS-LN	Perfluorooctane Sulfonic Acid - LN	1763-23-1-LN
PFOS-BR	Perfluorooctane Sulfonic Acid - BR	1763-23-1-BR
PFUnDA	Perfluoroundecanoic Acid	2058-94-8
PFNS	Perfluorononane Sulfonic Acid	68259-12-1
PFDoDA	Perfluorododecanoic Acid	307-55-1
PFDS	Perfluorodecane Sulfonic Acid	335-77-3
PFTrDA	Perfluorotridecanoic Acid	72629-94-8
FOSA	Perfluorooctane Sulfonamide	754-91-6
PFTeDA	Perfluorotetradecanoic Acid	376-06-7
11Cl-PF3OUdS	11-chloroeicosafluoro-3-oxaundecane-1-sulfonic acid	763051-92-9
9Cl-PF3ONS	9-chlorohexadecafluoro-3-oxanone1-sulfonic acid	756426-58-1
ADONA	4,8-dioxa-3H-perfluorononanoic acid	919005-14-4
HFPO-DA	Hexafluoropropylene oxide dimer	13252-13-6
FHpPA (7:3 FTCA)	3-Perfluoroheptyl propanoic acid	812-70-4
FPePA (5:3 FTCA)	3-Perfluoropentyl propanoic acid	914637-49-3
FPrPA (3:3 FTCA)	3-Perfluoropropyl propanoic acid	356-02-5
PFBSA	Perfluorobutanesulfonamide	30334-69-1
PFECHS	Perfluoro-4-ethylcyclohexanesulfonate	67584-42-3
PFHxSA	Perfluorohexanesulfonamide	41997-13-1

Analytical Laboratory Report

Sample Summary (1 samples)

Sample ID	Sample Tag	Matrix	Collected Date/Time
S60442.01	Biosolids - DIG.4	Sludge	04/01/24 14:15

Analytical Laboratory Report

Lab Sample ID: S60442.01
Sample Tag: Biosolids - DIG.4
Collected Date/Time: 04/01/2024 14:15
Matrix: Sludge
COC Reference: 159865

Sample Containers

#	Type	Preservative(s)	Refrigerated?	Arrival Temp. (C)	Thermometer #
1	15mL Centrifuge Tube	Trizma	Yes	1.3	IR
1	125mL Plastic	None	Yes	1.3	IR

Extraction / Prep.

Parameter	Result	Method	Run Date	Analyst	Flags
Initial wt. (g) / Final wt. (g) / Volume (ml)*	8.38/6.52/10	ASTM D7968-17M	04/10/24 07:00	SRP	

Inorganics

Method: SM2540B, Run Date: 04/02/24 12:54, Analyst: MAM

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Total Solids*	1.8	1		%	1		

Organics

34 PFAs, Method: ASTM D7968-17M, Run Date: 04/10/24 19:40, Analyst: KCV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
PFBA*	Not detected	4.5		ug/kg	299	375-22-4	X
PFPeA*	Not detected	1.5		ug/kg	299	2706-90-3	
4:2 FTSA*	Not detected	1.5		ug/kg	299	757124-72-4	
PFHxA*	3.3	1.5		ug/kg	299	307-24-4	
PFBS*	Not detected	1.5		ug/kg	299	375-73-5	
PFHpA*	Not detected	1.5		ug/kg	299	375-85-9	
PFPeS*	Not detected	1.5		ug/kg	299	2706-91-4	
6:2 FTSA*	Not detected	1.5		ug/kg	299	27619-97-2	
PFOA*	2.3	1.5		ug/kg	299	335-67-1	
PFHxS*	Not detected	1.5		ug/kg	299	355-46-4	
PFHxS-LN*	Not detected	1.5		ug/kg	299	355-46-4-LN	
PFHxS-BR*	Not detected	1.5		ug/kg	299	355-46-4-BR	
PFNA*	Not detected	1.5		ug/kg	299	375-95-1	
8:2 FTSA*	Not detected	1.5		ug/kg	299	39108-34-4	
PFHpS*	Not detected	1.5		ug/kg	299	375-92-8	
PFDA*	2.2	1.5		ug/kg	299	335-76-2	
N-MeFOSAA*	10	1.5		ug/kg	299	2355-31-9	
EtFOSAA*	2.1	1.5		ug/kg	299	2991-50-6	
PFOS*	Not detected	1.5		ug/kg	299	1763-23-1	
PFOS-LN*	Not detected	1.5		ug/kg	299	1763-23-1-LN	
PFOS-BR*	Not detected	1.5		ug/kg	299	1763-23-1-BR	
PFUnDA*	Not detected	1.5		ug/kg	299	2058-94-8	
PFNS*	Not detected	1.5		ug/kg	299	68259-12-1	
PFDoDA*	Not detected	1.5		ug/kg	299	307-55-1	
PFDS*	Not detected	1.5		ug/kg	299	335-77-3	
PFTrDA*	Not detected	1.5		ug/kg	299	72629-94-8	
FOSA*	Not detected	1.5		ug/kg	299	754-91-6	
PFTeDA*	Not detected	1.5		ug/kg	299	376-06-7	
11Cl-PF3OUdS*	Not detected	1.5		ug/kg	299	763051-92-9	
9Cl-PF3ONS*	Not detected	1.5		ug/kg	299	756426-58-1	

X-Elevated reporting limit due to matrix interference

Lab Sample ID: S60442.01 (continued)
Sample Tag: Biosolids - DIG.4

34 PFAs, Method: ASTM D7968-17M, Run Date: 04/10/24 19:40, Analyst: KCV (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
ADONA*	Not detected	1.5		ug/kg	299	919005-14-4	
HFPO-DA*	Not detected	1.5		ug/kg	299	13252-13-6	
FHpPA (7:3 FTCA)*	4.6	1.5		ug/kg	299	812-70-4	
FPePA (5:3 FTCA)*	48	1.5		ug/kg	299	914637-49-3	
FPrPA (3:3 FTCA)*	Not detected	1.5		ug/kg	299	356-02-5	
PFBSA*	Not detected	1.5		ug/kg	299	30334-69-1	
PFECHS*	Not detected	1.5		ug/kg	299	67584-42-3	
PFHxSA*	Not detected	1.5		ug/kg	299	41997-13-1	

Merit Laboratories Login Checklist

Lab Set ID: S60442
Client: MISCPFC (City of Cadillac - WWTP)
Project: Qtrly Biosolids
Submitted: 04/02/2024 10:00   Login User: MMC

Attention: Cindy Tomaszewski
Address: City of Cadillac - WWTP
1121 Plett Road
Cadillac, MI 49601

Phone: 231-775-2368   FAX:
Email: lab@cadillac-mi.net

Selection				Description	Note
Sample Receiving					
01.	☐ Yes	☒ No	☐ N/A	Samples are received at 4C +/- 2C Thermometer #	IR 1.3
02.	☒ Yes	☐ No	☐ N/A	Received on ice/ cooling process begun	
03.	☒ Yes	☐ No	☐ N/A	Samples shipped	UPS
04.	☐ Yes	☒ No	☐ N/A	Samples left in 24 hr. drop box	
05.	☒ Yes	☐ No	☐ N/A	Are there custody seals/tape or is the drop box locked	
Chain of Custody					
06.	☒ Yes	☐ No	☐ N/A	COC adequately filled out	
07.	☒ Yes	☐ No	☐ N/A	COC signed and relinquished to the lab	
08.	☒ Yes	☐ No	☐ N/A	Sample tag on bottles match COC	
09.	☐ Yes	☒ No	☐ N/A	Subcontracting needed? Subcontacted to:	
Preservation					
10.	☒ Yes	☐ No	☐ N/A	Do sample have correct chemical preservation	
11.	☐ Yes	☐ No	☒ N/A	Completed pH checks on preserved samples? (no VOAs)	
12.	☐ Yes	☒ No	☐ N/A	Did any samples need to be preserved in the lab?	
Bottle Conditions					
13.	☒ Yes	☐ No	☐ N/A	All bottles intact	
14.	☒ Yes	☐ No	☐ N/A	Appropriate analytical bottles are used	
15.	☒ Yes	☐ No	☐ N/A	Merit bottles used	
16.	☒ Yes	☐ No	☐ N/A	Sufficient sample volume received	
17.	☐ Yes	☒ No	☐ N/A	Samples require laboratory filtration	
18.	☒ Yes	☐ No	☐ N/A	Samples submitted within holding time	
19.	☐ Yes	☐ No	☒ N/A	Do water VOC or TOX bottles contain headspace	

Corrective action for all exceptions is to call the client and to notify the project manager.

Client Review By: ______________________ Date: __________

2680 East Lansing Dr., East Lansing, MI 48823
Phone (517) 332-0167 Fax (517) 332-4034
www.meritlabs.com

C.O.C. PAGE # 1 OF 1 159865

CHAIN OF CUSTODY RECORD

REPORT TO

CONTACT NAME: CINDY TOMASZEWSKI
COMPANY: CITY OF CADILLAC WWTP
ADDRESS: 1121 PLETT RD
CITY: CADILLAC STATE: MI ZIP CODE: 49601
PHONE NO.: 231-775-2368 CELL NO.: P.O. NO.:
E-MAIL ADDRESS: lab@cadillac-mi.net QUOTE NO.:

INVOICE TO

CONTACT NAME: ☒ SAME
COMPANY:
ADDRESS:
CITY: STATE: ZIP CODE:
PHONE NO.: E-MAIL ADDRESS:

PROJECT NO./NAME: QRTRLY BIOSOLIDS
SAMPLER(S) - PLEASE PRINT/SIGN NAME: DAVID SKIFF

TURNAROUND TIME REQUIRED ☐ 1 DAY ☐ 2 DAYS ☐ 3 DAYS ☒ STANDARD ☐ OTHER

DELIVERABLES REQUIRED ☐ STD ☒ LEVEL II ☐ LEVEL III ☐ LEVEL IV ☐ EDD ☐ OTHER

MATRIX CODE: W=WATER GW=GROUNDWATER WW=WASTEWATER S=SOIL L=LIQUID SD=SOLID SL=SLUDGE DW=DRINKING WATER O=OIL WP=WIPE A=AIR WS=WASTE

ANALYSIS (ATTACH LIST IF MORE SPACE IS REQUIRED)

Certifications: ☐ OHIO VAP ☐ Drinking Water ☐ DoD ☒ NPDES
Project Locations: ☐ Detroit ☐ New York ☐ Other

MERIT LAB NO. (FOR LAB USE ONLY)	COLLECTION DATE	COLLECTION TIME	SAMPLE TAG IDENTIFICATION-DESCRIPTION	MATRIX	# OF BOTTLES	NONE	HCl	HNO_3	H_2SO_4	NaOH	MeOH	OTHER	PFAS-MI	Special Instructions
60442.01	4-1-24	1415	BIOSOLIDS-DIG. 4	SL	4	X							X	ASTM D7968 (same as 559163.02)

RELINQUISHED BY: SIGNATURE/ORGANIZATION	DATE	TIME	RELINQUISHED BY: SIGNATURE/ORGANIZATION	DATE	TIME
CATomaszewski / City of Cadillac ☐ Sampler	4/1/24	1500	UPS	4/2/24	1000
RECEIVED BY: Ship via UPS Ground			RECEIVED BY: M Chilcote	4/2/24	1000
RELINQUISHED BY:			SEAL NO.: SEAL INTACT YES ☐ NO ☐ INITIALS:		
RECEIVED BY:			SEAL NO.: SEAL INTACT YES ☐ NO ☐ INITIALS:		

NOTES: TEMP. ON ARRIVAL 1.3

PLEASE NOTE: SIGNING ACKNOWLEDGES ADHERENCE TO MERIT'S SAMPLE ACCEPTANCE POLICY ON REVERSE SIDE

Rev. 1.26.22